

Question Paper

Exam Date & Time: 24-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

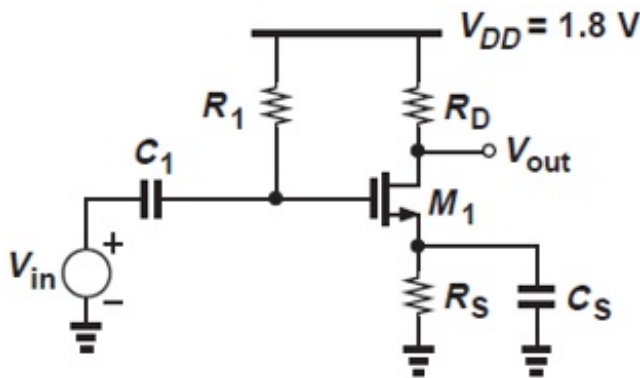
THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025 / JAN 2026
SUBJECT: ECE 2121/ECE_2121 ANALOG ELECTRONIC CIRCUITS
(SPL: ELECTRONICS AND COMMUNICATION ENGINEERING)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) The circuit shown in figure must provide a voltage gain of -6, with CS serving as a low impedance at the frequencies of interest. Assuming a power budget of 2 mW and an input impedance of 20 k Ω , design the circuit such that M1 operates 200 mV away from the triode region. Assume $\mu_n C_{ox} = 200 \mu\text{A/V}^2$, $\lambda=0$, and $V_{TH}=0.4\text{V}$, ignore capacitances. (4)

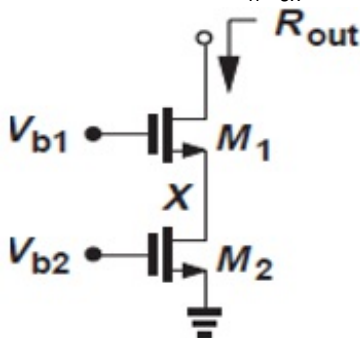


Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 1B) An NMOS device carries 1 mA with $V_{GS} - V_{TH} = 0.6 \text{ V}$ and 1.6 mA with $V_{GS} - V_{TH} = 0.8 \text{ V}$. If the device operates in the triode region, calculate V_{DS} and W/L . (3)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 1C) Design an NMOS cascode for an output impedance of 500 k Ω and a current of 0.5 mA. Assume M1 and M2 are identical, $\mu_n C_{ox} = 200 \mu\text{A/V}^2$ and $\lambda = 0.1\text{V}^{-1}$. (3)

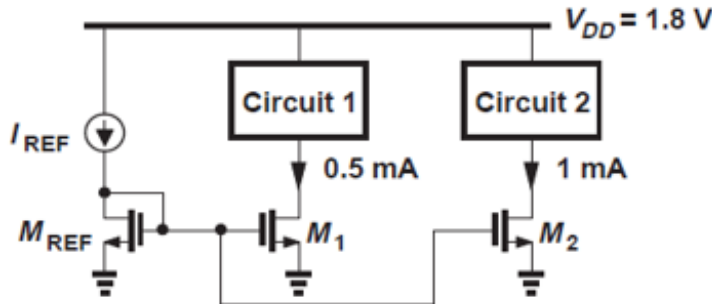


Course Outcome: "CO2" / Cognitive Level: "Application"

- 2A) Design an NMOS differential pair for a voltage gain of 5 and a power budget of 2 mW subject to the condition that the stage following the differential pair requires an input CM (common mode) level of at least 1.6 V. Assume $\mu_n C_{ox} = 100 \mu\text{A}/\text{V}^2$, $\lambda = 0$, and $V_{DD} = 1.8\text{V}$. (4)

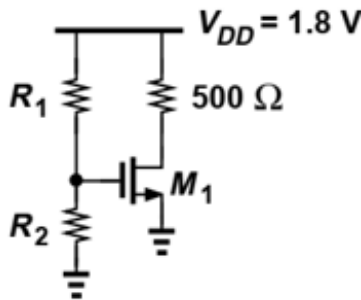
Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 2B) Figure shows an arrangement where M1 and M2 serve as current sources for circuits 1 and 2. Design the circuit for a power budget of 3 mW. (3)



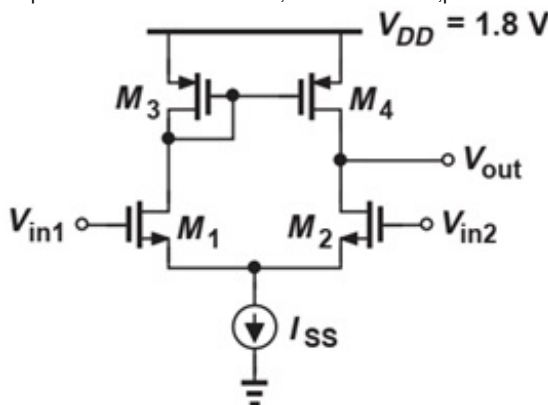
Course Outcome: "CO2" / Cognitive Level: "Application"

- 2C) Assume $\mu_n C_{ox} = 200 \mu\text{A}/\text{V}^2$, $\lambda = 0$, and $V_{TH} = 0.4\text{V}$. For a drain current of 1 mA. If $W/L = 20/0.18$. For the circuit shown, compute R1 and R2 such that the input impedance is at least 20K Ω . (3)



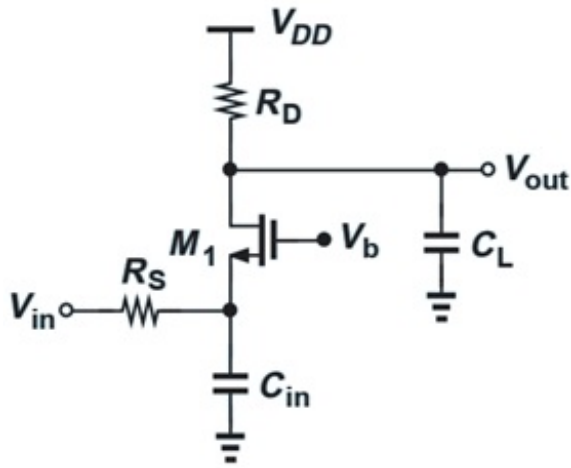
Course Outcome: "CO2" / Cognitive Level: "Application"

- 3A) Design the circuit shown below with a voltage gain of -20, power budget of 1 mW and $V_{DD} = 1.8\text{V}$. Assume M1 operates at the edge of saturation if the input common-mode level is 1V. Also, $\mu_n C_{ox} = 2\mu_p C_{ox} = 100 \mu\text{A}/\text{V}^2$, $V_{TH,n} = 0.5\text{V}$, $V_{TH,p} = -0.4\text{V}$, $\lambda_n = 0.5\text{V}^{-1}$, $\lambda_p = 0.1\text{V}^{-1}$. (4)



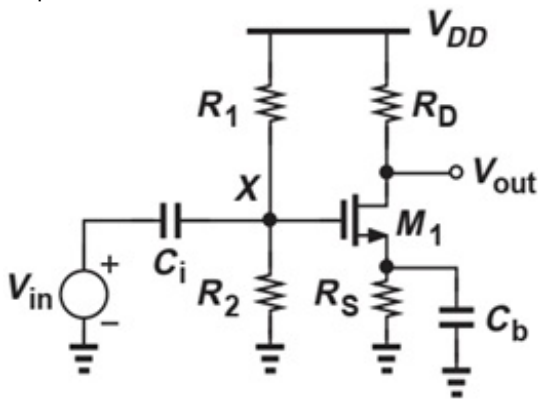
Course Outcome: "CO2" / Cognitive Level: "Application"

- 3B) For the amplifier shown, $g_m = (100\Omega)^{-1}$, $\lambda = 0$, $R_D = 2\text{K}\Omega$, $R_S = 200\Omega$, $C_{in} = 1\mu\text{F}$, and $C_L = 50\text{nF}$. Neglecting the MOSFET capacitance, sketch the frequency response of the amplifier indicating the pole frequencies. (3)



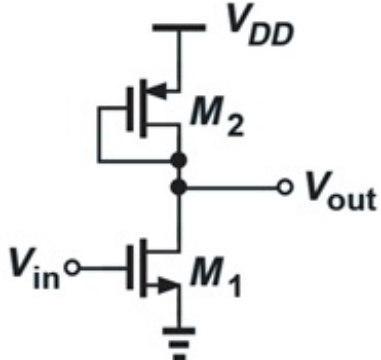
Course Outcome: "CO3" / Cognitive Level: "Application"

- 3C) For the circuit shown below, find the poles at low frequencies. Also, sketch its low frequency response. Assume $\lambda = 0$. (3)



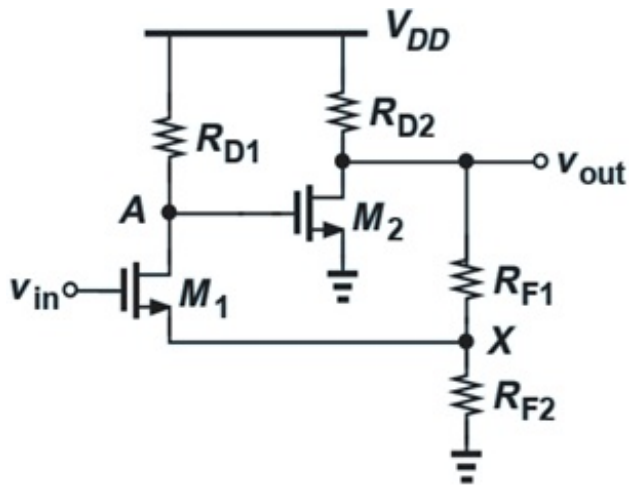
Course Outcome: "CO3" / Cognitive Level: "Application"

- 4A) For the circuit shown, determine the input and output poles at high frequencies for $\lambda = 0$. Sketch the frequency response. (4)



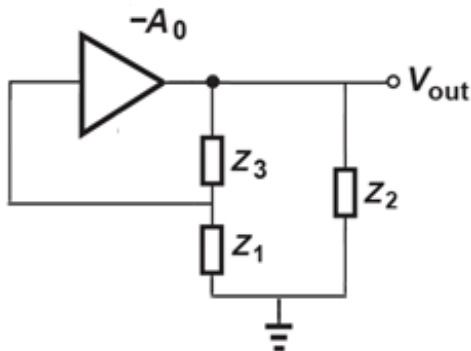
Course Outcome: "CO3" / Cognitive Level: "Application"

- 4B) Determine the polarity of feedback in the circuit shown below. (3)



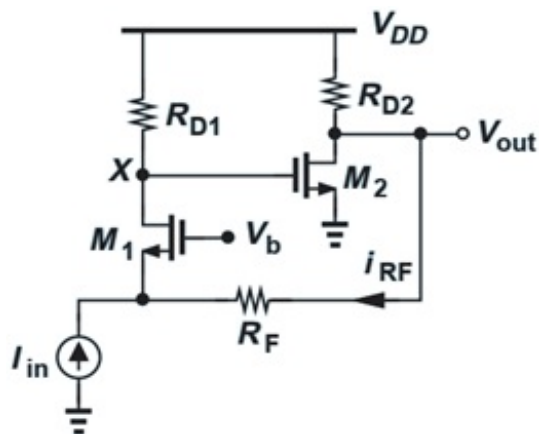
Course Outcome: "CO4" / Cognitive Level: "Understand"

- 4C) For the circuit shown below, obtain the expressions for gain and frequency of oscillation. Also, extend the result for Hartley and Colpitts's oscillator. (3)



Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5A) Determine the closed-loop gain, closed-loop input impedance, and closed-loop output impedance for the circuit shown below. Consider I/O effects (Loading Effect). Assume $\lambda=0$ for both M_1 and M_2 . (4)

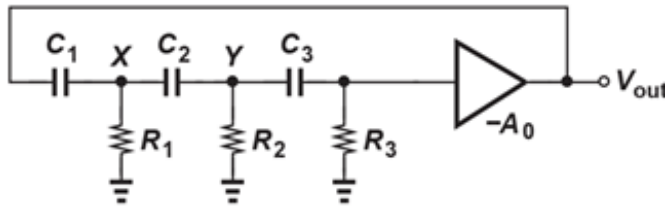


Course Outcome: "CO4" / Cognitive Level: "Application"

- 5B) Obtain an expression for % efficiency of Class A power amplifier and find its maximum value, also draw the circuit diagram. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5C) For the circuit shown below, obtain the expressions for gain and the frequency of oscillation. Also, find the frequency of oscillation for $R=100K\Omega$ and $C=0.1\mu F$. (3)



Course Outcome: "CO4" / Cognitive Level: "Understand"

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Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	ECE_2121:ANALOG ELECTRONIC CIRCUITS

#	Course Outcome	Description	Marks
1	1	CO 1	3
2	2	CO 2	21
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	6
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	ECE_2121:ANALOG ELECTRONIC CIRCUITS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	5	15(30%)
3	Application	8	27(54%)
4	Analysis	2	8(16%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)